Error Is Not Benefit: When Does Escalating to a Multi-Agent VLM Council Pay Off?

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Submission Id: 2056

Abstract

Multi-agent systems built from vision-language models (VLMs) can answer each input with one cheap call or with a council of specialist agents, and a budget forces the choice per input. Escalating pays off only if the council improves the inputs it receives, the router's signal finds those inputs, and both hold beyond the data used to build the system. We test these three conditions on surgical action-triplet recognition with a budget-aware gate, a council of phase, instrument, and triplet agents, two 8-bit backbones, and fifteen videos, ten of them drawn after every design decision. Error is not benefit: the cheap tier's uncertainty predicts its own mistakes (AUROC 0.66) but not which of them the council repairs (0.49), so an oracle that escalates only improved frames beats one that escalates every erroneous frame at half the cost, and the gate, routing on uncertainty, recovers only 3% of the gap between random escalation and that oracle. A coordination gain can be a label artifact: on the primary backbone the council beats the cheap tier on every development video and loses on every held-out video, and relabelling three few-shot exemplars, images unchanged, reversed the held-out result on every video, as predicted before the run. The only mechanism whose gain transferred passed the instrument agent's output to the triplet agent before it answered; filtering with the same output afterwards did nothing. Tracking a budget bounds spending but does not beat a calibrated threshold at matched cost. A screen that compares the cheap tier with a constant predictor on calibration data, fixed before the second backbone was rerun, correctly rejected the backbone on which gating failed. We distil four practices for evaluating cost-aware multi-agent systems.

Keywords

Multiagent coordination; Cost-aware inference; Budget-aware routing; Vision-language models; Evaluation methodology; Surgical video understanding

ACM Reference Format:
Anonymous Author(s). 2027. Error Is Not Benefit: When Does Escalating to a Multi-Agent VLM Council Pay Off?. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), Hanoi, Vietnam, 3–7 May 2027*, IFAAMAS, 9 pages.

1 Introduction

Multi-agent systems built from large language models let specialist agents generate, exchange, and combine intermediate outputs on tasks that a single call handles poorly [7, 9, 30]. Surgical video admits this decomposition: which instrument acts on which anatomical target, and in which phase, can each be assigned to an agent. But a council costs several calls per frame over thousands of frames, so a fixed budget forces a frame-by-frame decision.

Cost-aware routers make this decision for text queries [17, 31] and multi-agent workflows [14, 34]. Whatever the router, escalation pays off only under three conditions: **(C1)** the expensive tier scores higher on the inputs it receives; **(C2)** the router's signal identifies those inputs; **(C3)** both hold beyond the data on which the system was built. Learning to defer distinguishes the inputs a model gets wrong from the inputs an expert gets right [16, 19], and quality-gap routers train on the difference [6, 23]; both need labelled examples of the gap, which we did not collect (Section 6). Without them, routers from cascades [4] to confidence-thresholded selective prediction escalate on the cheap tier's own uncertainty, and thereby assume that C2 follows from predicting the cheap tier's error. This paper measures what that assumption costs when the expensive tier is a council, and tests the three conditions.

The task is action-triplet recognition in CholecT50 [21, 22]. The cheap tier is one call to a triplet agent; the expensive tier is a council of phase, instrument, and triplet agents combined by a rule-based critic and a fusion step; a budget-aware sequential gate, driven by the cheap tier's self-consistency uncertainty and a dual price tracking the budget left for the video, decides at each frame whether to escalate. We run two 8-bit backbones on fifteen videos, ten drawn at random after every design decision, with the video rather than the seed as the unit of analysis, and report, beside random, calibrated, and offline policies, an *error oracle* that escalates every frame the cheap tier gets wrong and a *benefit oracle* that escalates every frame the council improves; their gap is the price of the assumption.

Error is not benefit. The gate selects frames the cheap tier gets wrong more precisely than cost-matched random escalation (0.82 against 0.74), but the uncertainty that finds them, which predicts cheap-tier error with an AUROC of 0.66, separates the erroneous frames the council repairs from those it does not at 0.49. The benefit oracle reaches 0.377 Jac_IVT at 1.57 calls per frame, above the error oracle (0.344 at 3.25) and always-on council (0.284 at 4.00); the gate covers 13% of the distance from random escalation to the first ceiling and 3% to the second. The limit on escalation is the signal, not the budget or the council, on both backbones.

The council's advantage is a label confound. On the primary backbone the council beats the cheap tier on each of the five development videos and loses on each of the ten held-out videos. Three of its seven few-shot exemplars are labelled grasper-grasp-gallbladder, found in 30.1% of development frames and 1.4% of held-out frames, where grasper-retract-gallbladder occurs in 53.0%; relabelling those three, images unchanged, turned the council's loss on the held-out videos into a gain on every one, as predicted before the run. Three



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Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), M. Baldoni, F. Fang, W. Yeoh, N. Yorke-Smith (eds.), 3–7 May 2027, Hanoi, Vietnam. © 2027 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

labels move its advantage by 0.04 to 0.12 Jac_IVT, more than any coordination mechanism we ablate.

Coordination helped before the answer, not after it. Of the council's mechanisms, only passing the instrument agent's output to the triplet agent before it answers improved results, on every held-out video and at no extra cost. The fusion filter applies the same output after the triplet agent has answered and changes nothing; the critic, which also checks after the fact, changed no prediction in one round and hurt in two.

Budget tracking buys predictable spending, not accuracy. The gate stays within its cap by construction and, because the dual price rises as soon as spending exceeds the target rate, spreads escalations almost uniformly; a threshold on the same signal matches it at matched cost. A screen comparing the cheap tier with a constant predictor on calibration data, fixed before the second backbone was run at 8-bit precision, rejected that backbone, on which gating did not beat random escalation; a fixed-threshold screen admitted it.

The paper's contributions are the benefit oracle, the pre-specified intervention, the decomposition of the gate's gain, and the screen tested as a prediction. From them we distil four evaluation practices (Section 7): two we propose, a constant-predictor reference on calibration data and a benefit oracle beside the error oracle, and two we found necessary, evaluation on data drawn after every design decision and the video rather than the seed as the unit of analysis.

2 Related Work

Multi-agent LLM systems and cost. AutoGen [30], MetaGPT [9], and ChatDev [24] organize specialized agents through structured conversation, and debate has been reported to improve reasoning [7], although a single well-prompted agent is often competitive [27]; debate and critique act on answers already given, whereas Section 5.1 finds that the one mechanism that helped acts before the answer. None treats the number of calls as a constraint. Cost-aware routing does: SeqRoute [31] and drift-aware sparse routing [12] carry a budget across a sequence of requests, OmniRouter [17] optimizes over a query distribution under a global budget, ProgRouter [14] and Router-R1 [36] decompose one task into several routing steps, and MasRouter [34] and BAMAS [32], closest to our setting, jointly choose which LLMs collaborate and how, but once per task rather than frame by frame against a budget observed as it is spent. FrugalGPT [4] cascades a query through costlier models on the cheap model's confidence. None of this work routes agents that perceive images or video, and none reports evaluation on data drawn after all design decisions were fixed.

Error versus benefit. Learning to defer [16, 19] trains a classifier and a rejector jointly so that deferral depends on where the expert is right, not only on where the classifier is wrong; Hybrid LLM [6] and RouteLLM [23] apply the same principle to LLM routing by training on the quality gap between a small and a large model. Both require labelled examples of the gap. Selective classification [8], self-reported calibration [11], and self-consistency [29] instead supply signals that predict whether the model emitting them is wrong, and cascades that route on them [4] treat the two targets as one. We measure the cost of that substitution where no gap labels exist, with an error oracle and a benefit oracle, and screen each agent's confidence before admitting it to the council.

Exemplar sensitivity and surgical VLMs. Few-shot behaviour depends on which exemplars are shown and with what label balance: Zhao et al. [37] identify a majority-label bias, and Min et al. [18] find that label space and format often matter more than label correctness; in a multi-agent system such effects can masquerade as coordination gains (Section 5.1). For surgical video, PeskaVLP [33], SurgVLM [35], and SurgVidLM [26] pretrain or instruction-tune surgical VLMs, and SurgicalVLM-Agent [10] and SurgRAW [15] coordinate specialist agents at inference time; none reasons about cost. Supervised phase [5, 25] and triplet [20] methods remain the reference against which zero-shot results should be read.

3 Escalation Testbed

Figure 1 shows the testbed: a cheap tier, a council, a budget-aware gate between them, and diagnostics computed on calibration videos before any test video is seen. The gate is one instantiation of uncertainty-driven escalation under a budget.

3.1 Tiers and cost model

The single-agent tier π_{single} queries the triplet agent once at normalized cost $c_{\text{direct}} = 1$. The council queries a phase, an instrument, and a triplet agent at cost $c_{\text{council}} = 3$ per round and combines their outputs (Section 3.3); its triplet answer is obtained afresh under a different prompting configuration, not reused. A frame escalated for r rounds costs $c_{\text{direct}} + r\,c_{\text{council}}$, otherwise c_{direct}. Each video has its own budget $B = f \cdot T \cdot c_{\text{council}}$ at budget level f, so overspending on one cannot draw on another; T is the mean number of in-body frames per calibration video under the test-time sampling (Section 4.1).

3.2 Budget-aware sequential gate

Let $c_t \in [0, 1]$ be the self-consistency agreement of π_{single} on frame t, the mean pairwise Jaccard index among the triplet sets of five samples at temperature 0.7 (two empty sets count as agreeing), and $u_t = 1 - c_t$ its uncertainty.[1] The gate escalates if

$$u_t > \lambda_t \cdot c_{\text{council}} \quad \text{and} \quad B_t \geq c_{\text{council}} \cdot R, \tag{1}$$

where B_t is the budget remaining for the video, R the maximum number of council rounds a frame may consume, and λ_t a dual price. After observing the cost $s_t \in \{0, c_{\text{council}}, \ldots, R\,c_{\text{council}}\}$ incurred at t, the gate updates

$$\lambda_{t+1} = \text{clip}\Big(\lambda_t + \eta\big(s_t - \tfrac{B}{T}\big),\ 0,\ \lambda_{\max}\Big), \tag{2}$$

with target rate B/T, $\eta = 1/\sqrt{T}$, and $\lambda_{\max} = 10$. The price, reset to zero per video, moves by $\eta(c_{\text{council}} - B/T)$ on an escalation and $-\eta B/T$ otherwise, about +0.11 and −0.05 at budget 0.3, so it oscillates in a narrow band around the level that holds spending at the target rate: the price sets *how often* a frame may escalate and the uncertainty decides *which*. Since $u_t \leq 1$ and $c_{\text{council}} = 3$, Equation (1)

[1] The implementation also adds $\beta \cdot \mathbb{1}[\text{transition suspected at } t]$ for primary tasks that expose a transition signal; triplet recognition exposes none, so the term is zero throughout, and varying $\beta \in \{0.1, 0.3, 0.5, 0.7\}$ or $\lambda_{\max} \in \{5, 10, 20, 30\}$ at budget 0.3 reproduced the default escalation decisions frame by frame in all eight settings (Jac_IVT 0.282, 1.872 calls per frame).

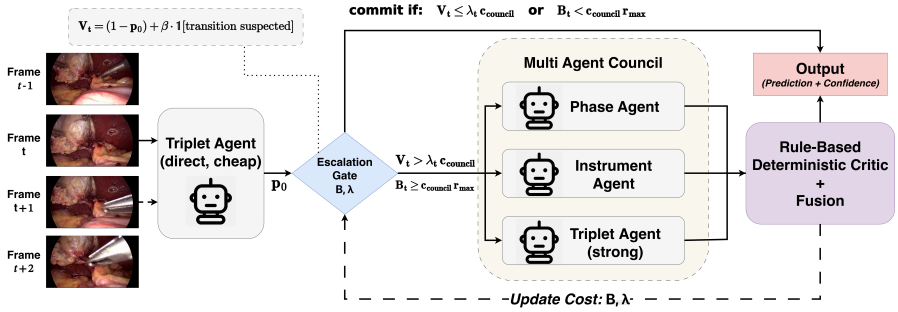


Figure 1: System overview. π_{single} emits a prediction and confidence c_t (p_0 in the figure); the gate escalates when uncertainty exceeds the dual price; the council's phase, instrument, and triplet agents feed a rule-based critic and fusion. $r_{\max}$ is the round limit R; the transition term is zero for the triplet task.

fires only while $\lambda_t < 1/3$, so the ceiling is never reached. The second condition admits a frame only if the remaining budget covers the most it can consume, so by induction a video's escalation cost never exceeds B, the only formal property we claim. The update follows dual mirror descent for online allocation [3], whose regret bounds assume costs in $[0, 1]$, a normalization we do not apply; we treat it as a pacing heuristic and evaluate it empirically.

3.3 Council and fusion

On escalation the phase, instrument, and triplet agents are queried independently. The instrument and triplet agents receive seven in-context exemplars, against four at π_{single}, and the triplet agent's answer is a majority vote over one greedy and five sampled decodings at temperature 0.7, keeping a triplet that appears in at least three of six ballots. The phase agent uses the four π_{single} exemplars and does not vote. All exemplars are calibration-video frames selected by manual review and fixed for every measurement; the council's seven add three to the four of π_{single} and were chosen during development, when the challenge-split videos were the only evaluation videos. Three of the seven, but one of the four, are labelled grasper-grasp-gallbladder (Section 5.1).

A rule-based critic, which makes no model call, flags a triplet whose instrument the instrument agent does not report as present and, when the phase agent's confidence exceeds 0.6, instruments implausible for the predicted phase or a phase inconsistent with those committed earlier in the video; at our main setting ($R = 1$) its verdict triggers no re-query, and a variant lets it re-query over a second round. A fusion step then removes any triplet whose instrument the instrument agent, at confidence above 0.6, does not report; fusion removes, never adds. In the *cross-conditioning* variant the instrument agent's prediction and confidence are passed to the triplet agent before it answers, as a hint the prompt marks as non-binding. Of the council's six mechanisms (exemplar set, voting, fusion, critic, admission by the screen below, cross-conditioning), only the last transfers information between agents before they answer.

3.4 Calibration-set diagnostics

Gating needs an informative confidence signal, and escalation has little to repair if the cheap tier is no better than a trivial policy; we test both on the calibration videos. *Agent-level screen, fixed thresholds.* For each agent we compute the AUROC with which each of three signals (token log-probability, self-consistency, verbalized confidence) predicts its own correctness, its top-1 accuracy, and the share of outputs taken by its most common prediction (degeneracy); it passes at best AUROC ≥ 0.70, accuracy ≥ 0.15, degeneracy ≤ 0.80, thresholds set during development. The signal-validated council drops a non-primary agent at degeneracy above 0.80 or best AUROC below 0.65; the triplet agent, which produces the output, is exempt. *Backbone-level screen, data-referenced.* The cheap tier must outperform the best constant prediction, the fixed set of up to three of the eight most frequent calibration triplets that scores highest, in accuracy (a frame is correct at Jac_IVT ≥ 0.5) and in Jac_IVT, each against the constant best under that metric, and its empty-prediction rate must be close to the empty-label rate; a backbone that fails is a candidate for always or never escalating, not for gating.

4 Evaluation Setup

Four research questions follow the conditions: **RQ1** (C1, C3), whether the council improves on the cheap tier, whether this transfers beyond development, and which mechanisms carry it; **RQ2** (C2), whether the cheap tier's uncertainty identifies the frames the council improves or only those the cheap tier gets wrong; **RQ3**, what budget tracking adds over random, calibrated, and offline policies at matched cost; **RQ4**, whether a calibration-set screen can anticipate the answers.

4.1 Data, video roles, and backbones

CholecT50 [22] contains 50 cholecystectomy videos annotated at 1 fps with 100 triplet classes over 6 instruments, 10 verbs, and 15 targets. Our agents are never trained, so we distinguish videos by whether they informed design decisions. Ten **calibration videos** (VID01, 02, 04, 06, 08, 10, 12, 13, 14, 15, public CholecT45 subset) supply the exemplars, the screens, the calibration of the threshold, conformal, and bandit baselines, and T. Five **challenge-split videos** (VID92, 96, 103, 110, 111), the CholecTriplet test split [21], were the only evaluation videos while prompts, exemplars, and gate hyperparameters were set. Ten **extended videos** (VID22, 32, 40, 47, 48, 62, 65, 66, 74, 75) were drawn uniformly at random with a fixed

seed from the 35 public videos not used for calibration, after every design decision, and were used for none. We sample 400 frames per test video at uniform intervals and drop out-of-body frames, identified from the image alone, leaving 5,727 frames; the same sampling fixes T and calibrates the baselines, while the screens use every annotated calibration frame (17,362 per agent). Two open 8-bit backbones are served with vLLM: the developers' FP8 release of Qwen3-VL-8B-Instruct [2] and InternVL3.5-8B [28] in a community FP8 build (brandonbeiler/InternVL3_5-8B-FP8-Dynamic, dynamic quantization from the official weights); an earlier 4-bit AWQ run serves only the screen comparison of Section 5.4. Prompts, exemplars, and hyperparameters are shared and were all developed on Qwen3-VL. The instrument agent decodes against a JSON schema; the triplet agent writes a short observation then a JSON list (prompts in the supplementary material).

4.2 Policies, references, and oracles

Every policy decides per frame from the cheap tier's output; only bandit also reads the phases committed earlier in the video. π_{single} never escalates; **always-on** always does. **Threshold** escalates when confidence falls below τ, the median cheap-tier confidence on the calibration videos. **Conformal** escalates when one-minus-confidence exceeds a split-conformal quantile [1] at miscoverage $\alpha = 0.1$. **Bandit** escalates when an offline logistic regression on uncertainty, phase entropy, and frames since the last phase change, trained on calibration videos to predict cheap-tier error, exceeds 0.5, an offline surrogate for a contextual bandit [13]. **Random-matched** escalates, per video and seed, a uniformly random set of frames of the gate's size.

The **constant predictor** outputs on every frame the set scoring highest in Jac_IVT on the calibration videos, {grasper-retract-gallbladder, hook-dissect-gallbladder}, at zero calls; it is the label prior against which every policy is read. Offline, **top-k** escalates the gate's number of frames, ranked by uncertainty over the whole video in hindsight, and **uniform-interval** the same number at evenly spaced positions; together they separate pacing from signal. The **error oracle** escalates every frame π_{single} gets wrong, the ceiling for any router that predicts cheap-tier error; the **benefit oracle** escalates every frame on which the council scores strictly higher, an upper bound on any frame-wise choice between the tiers, since any further escalation can only lower or leave the score. Offline references replay the π_{single} and always-on logs, which is exact only if a frame's council output does not depend on which earlier frames were escalated; since the critic reads previously committed phases we verified it: on Qwen3-VL every one of the 211,055 frames escalated by any of 26 gate configurations received exactly its always-on prediction and every one of the 350,191 non-escalated frames its π_{single} prediction, and on InternVL3.5 the same holds for 104,253 and 182,097 frames across 10 configurations. At budget 0.3 we ablate the council's mechanisms (Table 2), including a weak council whose triplet answer is the π_{single} answer, and Section 5.1 adds an intervention on exemplar labels at seed 0.

4.3 Metrics, cost, and statistics

Jac_IVT is the Jaccard index between predicted and ground-truth triplet sets averaged over frames, scoring 1.0 when both are empty; 9.4% of frames have empty ground truth, and the 7.1% that π_{single} also leaves empty contribute 0.071 of its 0.276 on Qwen3-VL (0.226 on non-empty frames). The convention shifts levels, not the differences we test. We do not report mAP, on which supervised methods score 4.2% to 38.1% [20]: our agents emit sets without per-class scores, and at our sampling mAP_IVT spans only 0.030 to 0.038 across every configuration. A call is one greedy plus five sampled generations sharing a prompt; the token ratio of π_{single} to always-on on Qwen3-VL (1:3.97) matches the call ratio (1:4), and latency scales with the escalation rate (1.08 s per frame for π_{single}, 2.85 at budget 0.3, 6.45 for always-on). Budget-aware and random-matched run at budgets 0.1, 0.2, 0.3, 0.5, 0.6, 0.65 on Qwen3-VL and 0.1, 0.3, 0.6 on InternVL3.5. Every configuration runs with five seeds that vary only inference-time sampling on fixed videos; the between-video standard deviation of Jac_IVT (0.049) is six times the between-seed one (0.008), so the unit of independence is the video: each contributes its mean over seeds and frames, configurations are compared by a two-sided Wilcoxon signed-rank test over fifteen videos, and comparisons repeated across budgets are Holm-corrected. Seed-level paired t-tests appear only for escalation precision, a property of the gate on fixed videos.

5 Results

Numbers are means over the fifteen test videos and five seeds unless stated, and tests are video-level (Section 4.3); where the challenge-split videos, on which the system was developed, and the extended videos diverge, we report both.

5.1 RQ1: Where does the council's benefit come from, and does it transfer?

The council's advantage depends on the video group. On Qwen3-VL the council beats the cheap tier on every challenge-split video, by 0.072 to 0.108, and loses on every extended video, by 0.005 to 0.063 (Figure 2, dark); over all fifteen the difference is +0.011 (+0.009 over frames) and not significant (five of fifteen, $p = 0.80$). On InternVL3.5 the council wins thirteen of fifteen (+0.094, $p = 0.0003$). The two groups annotate the same action differently: grasper-grasp-gallbladder occurs in 30.1% of challenge-split frames and 1.4% of extended frames, grasper-retract-gallbladder in 12.2% and 53.0%, and every video follows its group's preference. Three of the council's seven exemplars carry the grasp label, against one of the cheap tier's four, and the tiers inherit opposite preferences: the council predicts grasp on 29.2% and 30.8% of frames in the two groups and retract on 13.3% and 17.5%; the cheap tier, grasp on 13.6% and 14.2% and retract on 35.8% and 38.8%. By component (Jaccard over instrument, verb, and target sets; challenge, extended), the council is worse on instruments in both groups (0.685 → 0.654, 0.667 → 0.611), better on targets in both (0.421 → 0.450, 0.452 → 0.463), and better or worse on verbs with the group (0.428 → 0.470, 0.552 → 0.424); it predicts empty on 33.7% and 24.5% of frames against 14.2% and 8.0%, more often wrongly than rightly. InternVL3.5's council shares the grasp preference (39.0%, 41.6%) but beats its far weaker cheap tier (0.155; 49% to 60% empty) on every component in both groups, with a smaller verb gain on the extended videos (0.230 → 0.335) than on the challenge split (0.248 → 0.455).

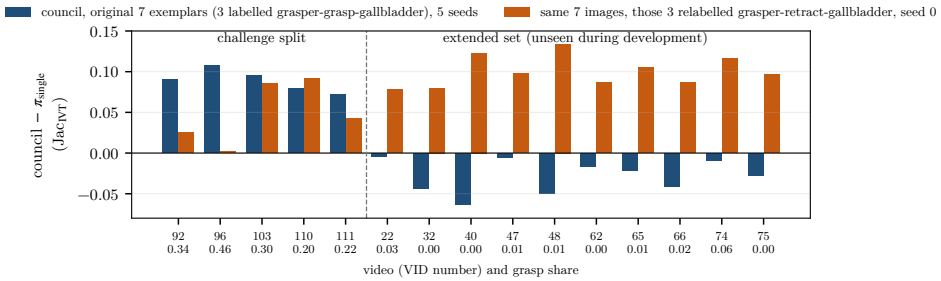


Figure 2: Council minus cheap tier, Jac_IVT per video, Qwen3-VL. Dark: original exemplars, mean of five seeds. Light: same images, the three grasp exemplars relabelled retract, seed 0, against the cheap tier of that run. Under each video: share of frames whose ground truth contains grasper-grasp-gallbladder.

Table 1: Relabelling intervention, Qwen3-VL, seed 0. Share of frames predicted to contain each triplet (challenge / extended) and Δ = council minus cheap tier Jac_IVT with videos won. Ground truth: grasp 0.301 / 0.014**, retract** 0.122 / 0.530**. Videos won at seed 0 (7/15) differ from the five-seed mean (5/15) in the text.**

	Cheap tier	Original council	Relabelled council
Predicts grasp-GB	0.141 / 0.144	0.300 / 0.313	0.027 / 0.024
Predicts retract-GB	0.355 / 0.385	0.131 / 0.177	0.400 / 0.477
Δ challenge (won)	–	+0.090 (5/5)	+0.050 (5/5)
Δ extended (won)	–	−0.024 (2/10)	**+0.100 (10/10)**
Δ all (won)	–	+0.014 (7/15)	+0.083 (15/15)

Intervention: the labels cause the dependence. We kept all seven exemplar images and their other triplets, relabelled the three grasp exemplars as retract, and reran the cheap tier and always-on council on all fifteen videos at seed 0, having first recorded four predictions: grasp predictions fall and retract predictions rise on every video; the council's deficit on the extended videos rises to near zero or above; its challenge-split advantage shrinks substantially; the gap between groups narrows. All four held (Table 1; Figure 2, light). Grasp predictions fell from about 30% of frames to under 3%; on the extended videos the council went from −0.024 (two of ten) to +0.100 (ten of ten); its challenge-split advantage shrank from 0.090 to 0.050; the group gap reversed from +0.114 to −0.051; over all videos it wins every one (+0.083, $p = 0.0001$). Regenerating the cheap tier within the run moves its per-video score by 0.008 on average, an order of magnitude below these effects. Three labels thus move the council's advantage by 0.04 to 0.12 Jac_IVT, more than any mechanism ablated below and ten times the gate's advantage over random escalation; where labels and annotation disagree, the margin is +0.050 (challenge split, relabelled) and −0.024 (extended, original), so the part of the advantage independent of agreement is small and of inconsistent sign. The relabelled council leaves the gate-versus-random comparison unchanged (both 0.300 at budget 0.3, eight of fifteen), consistent with Table 4; its labels were chosen after seeing the test videos, so it is a diagnostic, not a configuration.

Coordination helps before the answer, not after it. Table 2 ablates the six mechanisms at the gate's operating point. Cross-conditioning, which passes the instrument agent's prediction to the triplet agent as a non-binding hint, is the only one with a consistent benefit: +0.0132 Jac_IVT at unchanged cost, on thirteen of fifteen videos ($p = 0.0012$) and all ten extended videos. Part of the gain is a shift from grasp toward retract, but not all: with both grasper-gallbladder triplets removed from predictions and ground truth, it still improves Jac_IVT from 0.324 to 0.334 on the challenge split and 0.280 to 0.288 on the extended videos; the hint concerns instruments, the component on which the council trails the cheap tier in both groups. Signal-validated selection, which drops the phase and instrument agents on Qwen3-VL (Section 5.4), cuts tokens per frame by 23% and calls by 31% with no detectable change in Jac_IVT; on InternVL3.5 it drops only the non-voting phase agent (tokens −3.5%). The weak council and a second critic round have no detectable effect on Qwen3-VL, appearing harmful on the challenge split (−0.020, −0.016) and slightly helpful on the extended set, the group dependence of Figure 2; on InternVL3.5 a second round harms fourteen of fifteen videos (−0.017, $p = 0.0001$). Removing fusion has no detectable effect, although fusion applies the same instrument output that cross-conditioning passes, only after the triplet agent has answered; and the critic's phase-consistency check changed no prediction (Section 4.2). The one mechanism that transfers information between agents before they answer is thus the one that helps; those that check, filter, or admit independent outputs save cost without changing results; and three exemplar labels outweigh all of them (C1, C3).

5.2 RQ2: Does the signal find the frames the council improves?

The gate finds cheap-tier errors. On Qwen3-VL the gate's escalation precision against cheap-tier error exceeds cost-matched random escalation, whose precision equals the cheap tier's error rate of 0.748 in expectation, at every budget and in all thirty seed-by-budget comparisons (Table 5). The advantage survives on non-empty frames (0.811 to 0.838 against 0.794 to 0.801, $p \leq 0.03$ at every budget) but shrinks there to 0.014 to 0.037 of an aggregate gap of 0.067 to 0.094: most of it is the signal separating empty from non-empty frames. On InternVL3.5 the gate's precision sits at the error rate of 0.863. Offline top-k is more precise than the gate at every level on both backbones (0.825 to 0.850; 0.882 to 0.905).

Table 2: Ablations at budget-aware 0.3 (no fusion: against always-on): Jac_IVT mean ± std over five seeds, calls and tokens per frame, video-level difference from the reference (videos won, Wilcoxon p), and on the extended videos. Costs of the agent-exclusion rows are recomputed from the cost model.

Backbone	Configuration	Jac_IVT	Calls	Tokens	vs. reference (won, p)	Extended (won)
Qwen3-VL	Default budget-aware (0.3)	0.280 ± 0.002	1.872	9,736	–	–
	Signal-validated selection	0.281 ± 0.002	1.291	7,467	+0.0005 (9, 0.33)	+0.0020 (7)
	No instrument agent	0.281 ± 0.002	1.581	7,831	+0.0005 (9, 0.33)	+0.0020 (7)
	Cross-conditioning	**0.294 ± 0.001**	1.872	9,778	**+0.0132** (13, **0.0012**)	+0.0182 (10)
	One critic round ($R = 1$)	0.280 ± 0.002	1.872	9,736	identical to default	
	Two critic rounds ($R = 2$)	0.279 ± 0.002	1.872	9,117	−0.0015 (8, 0.93)	+0.0057 (8)
	Weak council	0.277 ± 0.001	1.872	6,823	−0.0039 (7, 0.45)	+0.0041 (7)
	No fusion (vs. always-on)	0.286 ± 0.003	4.000	20,733	+0.0020 (9, 0.42)	+0.0031 (7)
InternVL3.5	Default budget-aware (0.3)	0.182 ± 0.008	1.889	17,630	–	–
	Signal-validated selection	0.182 ± 0.008	1.889	17,019	identical predictions	
	Two critic rounds ($R = 2$)	0.165 ± 0.004	1.887	15,959	**−0.0170** (1, **0.0001**)	−0.0152 (1)

Table 3: Jac_IVT and cost (calls per frame), mean over five seeds, over all fifteen videos and by group. Budget rows give budget-aware / random-matched at equal cost. The constant predictor makes no call; the error oracle escalates whenever the cheap tier is wrong, the benefit oracle whenever the council scores higher on that frame.

	Qwen3-VL				InternVL3.5			
Configuration	All	Challenge	Extended	Cost	All	Challenge	Extended	Cost
Constant predictor	0.292	0.144	0.360	0	0.292	0.144	0.360	0
π_{single}	0.276	0.256	0.285	1.000	0.155	0.190	0.139	1.000
Budget 0.1 (BA / RM)	0.279 / 0.276	0.271 / 0.264	0.283 / 0.282	1.296	0.166 / 0.164	0.208 / 0.203	0.146 / 0.147	1.297
Budget 0.2 (BA / RM)	0.280 / 0.278	0.280 / 0.274	0.280 / 0.280	1.588	–	–	–	–
Budget 0.3 (BA / RM)	0.280 / 0.277	0.291 / 0.282	0.276 / 0.275	1.872	0.182 / 0.183	0.231 / 0.233	0.159 / 0.160	1.889
Budget 0.5 (BA / RM)	0.284 / 0.278	0.308 / 0.298	0.272 / 0.269	2.431	–	–	–	–
Budget 0.6 (BA / RM)	0.284 / 0.279	0.314 / 0.307	0.270 / 0.266	2.694	0.204 / 0.210	0.271 / 0.276	0.174 / 0.180	2.726
Budget 0.65 (BA / RM)	0.286 / 0.280	0.319 / 0.311	0.270 / 0.266	2.818	–	–	–	–
Conformal	0.277	0.262	0.285	1.279	0.156	0.191	0.139	1.012
Threshold	0.284	0.288	0.282	2.205	0.193	0.246	0.169	1.941
Bandit	0.289	0.315	0.276	2.849	0.207	0.273	0.177	2.556
Always-on council	0.284	0.344	0.257	4.000	0.248	0.334	0.208	4.000
Error oracle	0.344	0.374	0.330	3.245	0.295	0.386	0.253	3.589
Benefit oracle	**0.377**	0.392	0.370	1.568	**0.312**	0.391	0.276	1.852

Table 4: AUROC of the cheap tier's uncertainty for predicting, per frame, that the cheap tier is wrong, that the council scores higher, that it scores lower, and, among frames the cheap tier gets wrong, that the council scores higher. Relabelled: the intervention of Section 5.1, seed 0.

	Wrong	Helps	Harms	Fixable
Qwen3-VL, original council	0.663	0.541	0.513	**0.485**
Qwen3-VL, relabelled council	0.663	0.540	0.529	0.479
InternVL3.5	0.569	0.536	0.523	**0.525**

But error is the wrong target. Escalation helps only where the council scores higher: on Qwen3-VL, 18.9% of frames, while 18.6% are made worse and 62.5% are unchanged. The cheap tier's uncertainty predicts its own error with an AUROC of 0.663 but council improvement at 0.541, council harm at 0.513, and, among the frames the cheap tier gets wrong, repair at 0.485 (Table 4). The values are similar for the relabelled council, whose average benefit is larger, and for InternVL3.5, whose council is far stronger than its cheap tier, so this is not an artifact of a council that barely helps. The other cheap signals we log (uncertainty, number of predicted triplets, empty prediction) correlate with cheap-tier error at 0.277, 0.338, −0.355 and with council improvement at 0.038, −0.024, −0.004.

The oracle rows of Table 3 bound the consequence. The benefit oracle reaches 0.377 at 1.568 calls per frame on Qwen3-VL, above the error oracle (0.344 at 3.245) and always-on (0.284 at 4.000); at budget 0.1 it already reaches 0.351, above always-on at a third of its calls. At the gate's escalation count at budget 0.3, oracular knowledge of which frames are wrong yields 0.303 and of which frames the council improves 0.377; the gate, at 0.280 against random's 0.277, covers 13% of the distance to the first ceiling and 3% to the second. This is the paper's central measurement on C2: the gate's precision is precision on the wrong target, since the council repairs about one in four of the erroneous frames it finds and the signal carries almost no information about which; routing on cheap-tier error, as confidence cascades do, forfeits most of the attainable gain. The oracle is an upper bound in a further sense: it selects frames on which the council scored higher in the same stochastic run, so part of its margin is selection on sampling noise, and the near-symmetry of frames helped (18.9%) and harmed (18.6%) suggests that part is not small.

5.3 RQ3: What does budget tracking add?

Three features of Table 3 frame this section. The constant predictor scores 0.292 over all fifteen videos, above every deployable configuration on Qwen3-VL: the label shift of Section 5.1 acting on the prior, so the comparisons below concern policies on the same agents, not the task. Always-on is not reliably better than the cheap tier on Qwen3-VL (0.284 against 0.276, opposite signs by group) but far better on InternVL3.5. The gate's realised escalation rates are 0.099, 0.196, 0.291, 0.477, 0.565, 0.606; threshold and bandit escalate 40.2% and 61.6% of test frames unconstrained.

Against random escalation. On Qwen3-VL the gate's Jac_IVT exceeds cost-matched random at every budget (Table 5) but survives Holm correction only at 0.65 (+0.0059, twelve of fifteen, adjusted $p = 0.032$); at the operating point 0.3 it is +0.0035 ($p = 0.15$), and almost all of the low-budget advantage comes from the challenge split, the gate winning five of ten extended videos at budgets 0.1 to 0.3 by under 0.001. On InternVL3.5 the gate never beats random and is significantly worse at 0.6 (two of fifteen, adjusted $p = 0.006$).

Against calibrated baselines. At matched cost the gate is no more accurate than routers that track no budget. On Qwen3-VL threshold matches budget-aware 0.5 (0.284 against 0.284, eight of fifteen, $p = 0.98$) at 9% lower cost, bandit edges budget-aware 0.65 at almost the same cost (+0.0027, eleven of fifteen, $p = 0.23$), and budget-aware 0.65 equals always-on (+0.0007, $p = 0.98$) at 70% of its cost. On InternVL3.5 threshold beats budget-aware 0.3 on fourteen of fifteen videos (+0.0113, $p = 0.0001$) at 3% higher cost, and budget-aware 0.6 trails always-on by 0.044 (one of fifteen, $p = 0.0002$).

Decomposition and timing. Table 5 splits the gate's gain over the cheap tier into escalating at all (spending), spreading escalations evenly (spacing), and choosing them by confidence (signal). On Qwen3-VL the spending term is small, the spacing term lies between −0.0016 and +0.0005, and the signal term carries most of the gain; on InternVL3.5 spending carries it and the signal term lies between −0.0044 and +0.0011. On neither backbone does the gate beat offline top-k; on InternVL3.5 at 0.3 it trails on fourteen of fifteen videos. Timing explains why. At budget 0.3 on Qwen3-VL the gate places 19/21/21/21/19% of its escalations in the five fifths of a video, the profile of random escalation, while threshold places 15% rising to 24%, following the uncertainty (threshold precision 0.836 and 0.909 against the gate's 0.822 and 0.873 on the two backbones). Because the price rises as soon as spending exceeds the target rate, the gate escalates frames on average less uncertain than top-k's (0.740 against 0.826) on a more regular schedule (standard deviation of the gap between escalations 1.75 against 3.28 frames). The dual price is a rate limiter: it makes spending predictable, which a procedure-level cost cap needs, and it is why the gate cannot beat the offline ranking of its own signal.

5.4 RQ4: Can a calibration-set screen anticipate this?

Could the calibration videos alone have anticipated the results above (Table 6)? Under the fixed thresholds only the triplet agent passes on Qwen3-VL, narrowly (0.704 against 0.70); the instrument agent fails on AUROC despite its accuracy of 0.762, so confidence reliability and accuracy are separate properties, and much of the triplet agent's discrimination is the empty-versus-nonempty distinction (AUROC 0.580 on frames with a triplet). On InternVL3.5 the fixed thresholds admit the triplet agent, on a verbalized-confidence AUROC of 0.840 parseable on 1.5% of frames and an accuracy that clears the threshold by 0.005; gating on this backbone does not beat random escalation, so the verdict is a false positive.

The data-referenced screen uses {grasper-retract-gallbladder} as the best constant in accuracy and {grasper-retract-gallbladder, hook-dissect-gallbladder} in Jac_IVT. InternVL3.5 fails every variant: below both constants, behind the Jac_IVT constant on seven of ten calibration videos, and empty 4.66 times as often as the labels warrant. Qwen3-VL sits on the boundary: above the Jac_IVT constant on seven of ten videos, with an empty rate matching the labels (11.5% against 11.6%), but short on accuracy. The outcomes agree: on InternVL3.5 gating does not beat random; on Qwen3-VL it improves frame selection but end-task quality only marginally.

We fixed the accuracy criterion after the first runs of both backbones, when InternVL3.5 was served at 4-bit precision, and before its 8-bit rerun. Its 4-bit verdict is therefore not evidence. Its 8-bit verdict preceded the outcome, holds for every candidate constant, and reverses the fixed thresholds' own verdict. We then searched fewer constants, under which Qwen3-VL passed on accuracy; the complete search places it on the boundary, and the Jac_IVT comparison was added afterwards. The 4-bit run (triplet agent AUROC 0.743, accuracy 0.132, degeneracy 0.790, empty on 79.0% of calibration frames against 54.3% at 8-bit) shows why the rerun was needed: quantization explained much of the degeneracy, but at 8-bit the agent still trails the constant. The screen is informative only insofar as the calibration videos represent the test videos, where the same constant outscores Qwen3-VL's cheap tier, and it tests the cheap tier's signal, not whether the council repairs what the signal flags. As a case study, it delivered one correct prediction where a fixed-threshold screen erred and one boundary verdict that Section 5.3 bears out; it is a practice, not a validated instrument.

6 Limitations

Scope. One dataset, fifteen test videos at 400 frames each, two backbones of similar size, and zero-shot agents that do not compete with supervised methods. Our claims concern policies and mechanisms compared on the same agents, not surgical action recognition itself. The contrast between the extended and challenge-split videos would be strengthened by further held-out videos, in particular from the recommended cross-validation split [21], and frame strata such as phase transitions are too sparse for stratified analysis. **The label shift.** We show that exemplar labels determine which group the council wins, not why the groups annotate the grasper's action differently: the two verbs differ in intent and motion rather than in a single frame, so the shift may be an annotation convention or a real difference between procedures (the calibration videos are mixed, five favouring each verb). The intervention, on one seed and one backbone, establishes the direction of causation, not the effect size on other exemplar sets; its four predictions are the only prespecified element of the paper, the rest of the analysis having been designed after the held-out results were seen. Cross-conditioning was evaluated at one budget on one backbone. **The screen** has one genuine prediction; a benefit-referenced screen would need

Table 5: Budget-aware gate (BA) against random-matched (RM) and offline top-k at each budget: escalation precision (mean ± std over seeds); video-level Jac_IVT of BA − RM (mean, videos won, Wilcoxon p / Holm-adjusted across budgets, and per group); the decomposition of BA − π_{single} (total) into spending (RM − π_{single}), spacing (uniform-interval − RM), and signal (BA − uniform-interval); and BA − top-k (won, p). Recall and F1 are in the supplementary material.

		Precision		BA − RM, Jac_IVT				BA − π_{single}, Jac_IVT				
Backbone	Budget	BA	RM	Diff. (won)	p / Holm	Challenge	Extended	Total	Spending	Spacing	Signal	BA − top-k (won, p)
Qwen3-VL	0.1	0.842 ± 0.003	0.748 ± 0.018	+0.0029 (9)	0.135 / 0.406	+0.0070 (4)	+0.0008 (5)	+0.0033	+0.0005	−0.0011	+0.0038	+0.0012 (8, 0.60)
	0.2	0.823 ± 0.007	0.743 ± 0.017	+0.0026 (10)	0.252 / 0.406	+0.0066 (5)	+0.0005 (5)	+0.0044	+0.0020	−0.0007	+0.0031	+0.0002 (6, 0.85)
	0.3	0.822 ± 0.004	0.741 ± 0.007	+0.0035 (10)	0.151 / 0.406	+0.0089 (5)	+0.0008 (5)	+0.0048	+0.0015	−0.0006	+0.0039	−0.0025 (5, 0.28)
	0.5	0.816 ± 0.002	0.742 ± 0.004	+0.0055 (12)	0.013 / 0.062	+0.0098 (5)	+0.0033 (7)	+0.0078	+0.0025	−0.0016	+0.0069	−0.0029 (7, 0.42)
	0.6	0.815 ± 0.004	0.744 ± 0.005	+0.0051 (11)	0.022 / 0.086	+0.0070 (5)	+0.0041 (6)	+0.0085	+0.0035	+0.0004	+0.0046	−0.0047 (6, 0.25)
	0.65	0.814 ± 0.004	0.747 ± 0.005	+0.0059 (12)	0.005 / **0.032**	+0.0084 (4)	+0.0046 (8)	+0.0100	+0.0042	+0.0005	+0.0053	−0.0035 (6, 0.28)
InternVL3.5	0.1	0.867 ± 0.032	0.854 ± 0.008	+0.0013 (9)	0.525 / 0.842	+0.0055 (5)	−0.0008 (4)	+0.0104	+0.0093	+0.0001	+0.0011	+0.0000 (8, 0.93)
	0.3	0.873 ± 0.020	0.857 ± 0.006	−0.0010 (6)	0.421 / 0.842	−0.0017 (1)	−0.0007 (5)	+0.0268	+0.0278	+0.0002	−0.0012	−0.0072 (1, **0.0002**)
	0.6	0.864 ± 0.022	0.863 ± 0.004	−0.0059 (2)	0.002 / **0.006**	−0.0049 (1)	−0.0063 (1)	+0.0492	+0.0551	−0.0015	−0.0044	−0.0036 (5, 0.07)

Table 6: Screens on the ten calibration videos and the test outcome they are meant to predict. Agent rows list phase / instrument / triplet; best signal is self-consistency, except log-probability (InternVL3.5 phase) and verbalized confidence (InternVL3.5 triplet, parseable on 1.5% of frames; self-consistency AUROC 0.734). Constants are the best fixed answers on the calibration videos under each metric. Outcomes at budget 0.3: escalation precision minus the cheap tier's error rate, and BA minus RM Jac_IVT (videos won).

	Qwen3-VL	InternVL3.5
Agent screen (AUROC ≥ 0.70, acc. ≥ 0.15, degen. ≤ 0.80)		
Best AUROC	0.522 / 0.530 / 0.704	0.555 / 0.700 / 0.840
Accuracy	0.308 / 0.762 / 0.249	0.257 / 0.803 / 0.155
Degeneracy	0.736 / 0.359 / 0.115	0.332 / 0.448 / 0.543
Verdict	fail / fail / pass	fail / pass / pass
Backbone screen (data-referenced), triplet agent		
Accuracy: cheap / constant	0.249 / 0.262	0.155 / 0.262
Jac_IVT: cheap / constant	0.266 / 0.245	0.174 / 0.245
Videos above constant	7/10	3/10
Empty predictions / labels	0.98	4.66
AUROC: all / non-empty frames	0.704 / 0.580	0.734 / 0.638
Verdict	boundary	**fail**
Test: precision − error rate	+0.074	+0.010
Test: BA − RM (won)	+0.0035 (10/15)	−0.0010 (6/15)

council outputs on the calibration videos, which we did not collect. **The second backbone** runs from a community 8-bit build with prompts developed on Qwen3-VL, so its failure to benefit from gating may partly reflect prompts tuned to another model; 8-bit precision cut its empty-prediction rate from 79% to 54%, and higher precision, beyond our hardware, might cut it further. **Gate and critic.** The step size η binds and was not swept; a larger step makes the gate more like a threshold, a smaller one more like uniform escalation, and neither would close the gap to the benefit oracle. The critic's instrument-phase table was written by us following the phase definitions of [25], not reviewed by a surgeon, and at $R = 1$ cannot change a prediction. **Predicting benefit.** The signals we log correlate with cheap-tier error at 0.28 to 0.36 in absolute value and with council benefit at 0.04 or less; a predictor would need signals of another kind, such as disagreement between two cheap heterogeneous agents.

7 Conclusion

We asked when escalating a frame from one VLM call to a council of specialist agents pays off and tested the three conditions under which it can. On the primary backbone the council's advantage over the cheap tier held on the development videos and reversed on every held-out video, and a pre-specified relabelling of three few-shot exemplars reversed it back, outweighing every coordination mechanism; only passing the instrument agent's output to the triplet agent helped on every held-out video. The cheap tier's uncertainty predicted its own error but not the frames the council improves, on both backbones: error is not benefit, and routing on the former forfeits most of the attainable gain. Budget tracking bounded spending but did not beat a calibrated threshold at matched cost, and a calibration-set screen with a constant-predictor reference, fixed before the second backbone's 8-bit rerun, rejected that backbone, on which gating failed, where a fixed-threshold screen admitted it.

Four practices follow for evaluating cost-aware multi-agent systems: compare the cheap tier with references computed on calibration data, including a constant predictor; report a benefit oracle beside the error oracle; evaluate on data drawn after development and compare the label distribution of in-context exemplars with it; and use the video, or an equivalent independent unit, rather than the seed as the unit of analysis. One design observation follows as well: passing one agent's output to another before it answers can help where checking or filtering independent outputs does not.

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